# Customer Churn Prediction Report

Telco Dataset    April 2026    Prepared for management review

01 BUSINESS QUESTION

## What problem did we solve?

Each month, a portion of your customers cancel their service. This is called churn. Acquiring a new customer costs more than keeping an existing one. You need to know which customers are about to leave before they actually do. This project builds a system that reads each customer's profile and assigns them a churn probability score. Your retention team can then contact the highest-risk customers before they walk out.

> The core question: Which customers are most likely to cancel in the next billing cycle, and what do they have in common?

02 THE DATA

## What we analyzed

| Total customers | Churn rate | Variables per customer |
|---|---|---|
| 7,043 | 26% | 20 |
| After cleaning | 1,869 customers left | Raw, plus 6 engineered |

The dataset covers demographics, contract details, billing information, and service subscriptions. We removed 11 rows with missing billing data (0.15% of the total). We then built 6 additional variables from the existing ones to give the model a clearer picture of each customer.

03 WHAT THE DATA SHOWS

## The five biggest churn drivers

- Contract type is the strongest predictor. Customers on month-to-month contracts churn at 42%. Customers on two-year contracts churn at less than 5%. This one variable explains more about churn than anything else in the dataset.
- New customers leave the most. The first 12 months carry the highest churn rate. After two years, the rate drops sharply. Early engagement and onboarding are your best retention tools.

- Fiber optic customers churn at 41%. This is your premium internet segment. Higher monthly charges and stronger competition in that category appear to drive their departure.
- Electronic check is the payment method with the highest churn. These customers churn at 45%. This may reflect financial friction or lower engagement with the brand overall.
- Add-on services keep customers. Customers with online security, tech support, or device protection churn at a much lower rate. Each additional service a customer holds reduces their likelihood of leaving.

04 THE MODELS

## How the prediction works

We built two separate models and compared them. Each model reads a customer's profile and outputs a number between 0 and 100. That number is the probability that the customer will churn. We then flag anyone above a set cutoff as at risk.

Selected

### Logistic Regression

This model assigns a weight to each customer attribute. It adds those weights together and produces a probability score. You can trace exactly why any customer received a high or low score. It is fast, transparent, and easy to explain to non-technical teams. It scored higher on every key metric in this project.

Comparison

### Random Forest

This model runs 200 independent decision trees on each customer, then takes a majority vote. It finds complex patterns that a single model misses. It is harder to explain to stakeholders. In this project, it scored slightly lower than Logistic Regression on both recall and AUC.

Both models trained on 80% of customers. We tested them on the remaining 20%, which is data neither model had seen. This gives you an honest picture of how the model performs in real conditions.

05 PERFORMANCE METRICS

## How to read the results

Standard accuracy is misleading for this type of problem. A model that predicts no one churns would be 74% accurate and catch zero at-risk customers. We use four measures that actually reflect business value.

1. ROC-AUC (0.84). This measures how well the model ranks customers by risk. A score of 0.84 means the model correctly identifies a churner above a non-churner 84% of the time when comparing two random customers. Scores above 0.80 are considered strong.
2. Recall (0.72). Of all the customers who actually churned, the model caught 72% of them. This is the most important number for your retention team. A missed churner is a lost customer with no chance to intervene.
3. Precision (0.55). Of every customer the model flagged as at risk, 55% actually churned. The other 45% received a retention offer they did not need. Since outreach costs are low compared to losing a customer, this rate is acceptable.
4. Threshold (0.35). By default, models flag customers only when their score exceeds 0.50. We lowered this to 0.35. The model catches more churners at a slight cost to precision. In a retention context, catching more at-risk customers is the right trade.

| Metric | Before Engineering | After Engineering | Change |
|---|---|---|---|
| ROC-AUC | 0.8357 | 0.8372 | +0.0015 |
| Churn Recall | 0.72 | 0.72 | Stable |
| Churn Precision | 0.55 | 0.55 | Stable |
| Churn F1 | 0.62 | 0.62 | Stable |

06 PREDICTION OUTCOMES

## What the model did with 1,407 test customers

Every prediction falls into one of four categories. Two are correct. Two are errors. The two errors carry very different costs.

Correctly retained

916

Customer stayed. Model said they would stay. No action needed.

False alarm

117

Customer stayed, but model flagged them as at risk. They receive an unnecessary offer. Low cost.

Missed churner

159

Customer left, but model did not flag them. No intervention was made. High cost.

Correctly caught

215

Customer was going to leave. Model flagged them. Retention team can act.

Out of 374 customers who actually churned in the test set, the model caught 215. Without any model, you catch zero. After threshold tuning, the model recovers approximately 54 additional churners compared to the default setting.

07 FEATURE ENGINEERING

## The six variables we built from scratch

These variables do not exist in the original data. We created them by combining raw columns to surface patterns the model could not detect on its own.

| Variable | What it measures | Why it matters |
|---|---|---|
| `tenure_group` | Bins tenure into 4 life stages | New customers behave differently from loyal ones. A number alone does not capture that. |
| `num_services` | Count of active subscriptions | Each additional service raises the cost of switching. More services means less churn. |
| `avg_monthly_spend` | Total charges divided by tenure | Finds customers paying more than their history justifies. A friction indicator. |
| `is_high_value` | Monthly charges above median | Premium customers may need a different retention approach than standard ones. |
| `at_risk_combo` | Month-to-month plus electronic check | Combines the two strongest individual signals into one direct flag. |
| `has_security_addon` | Any security service active | Security add-ons are a retention anchor. Their absence is a warning sign. |

AUC improved from 0.8357 to 0.8372 after adding these features. The recall held steady at 0.72. These variables are also built into the Power BI dashboard as filters so your team can segment customers without reading raw scores.

08 NEXT STEPS

## What you can do with this today

### Target the at-risk combo group

Month-to-month customers paying by electronic check in their first 12 months. This group has the highest observed churn rate in the entire dataset.

### Promote add-on services early

Each service a customer adds reduces their churn probability. Offer security and support add-ons within the first three months of onboarding.

### Review fiber optic pricing

### Refresh the model monthly

41% churn rate in a premium segment is a pricing or competition issue worth investigating before the next quarter.

Retrain the model as new customer data arrives. Customer behavior shifts over time and the model should reflect current patterns.

> All predictions, risk scores, and customer segments are exported to churn_dashboard_data.csv and ready to load into Power BI. Each customer carries a probability score, a binary prediction, and a risk label of Low, Medium, or High.